

System Understanding Document (SUD)

Project Name: PwC SAP S/4HANA Migration - Cloud Infrastructure & Automation

Document Type: System Architecture & Functional Overview

Target Audience: Engineering Leads, Cloud Architects, DevOps Engineers, and Onboarding Personnel

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1. Executive Summary & Core Objectives

This System Understanding Document (SUD) provides a comprehensive technical breakdown of the **PwC SAP S/4HANA Migration - Cloud Infrastructure & Automation** platform. The ecosystem bridges legacy Enterprise Resource Planning (ERP) databases with modern cloud infrastructure, integrating AI-driven invoice automation, scalable backend microservices, real-time API webhooks, and enterprise-grade observability.

Key Functional Deliverables

- **SAP S/4HANA Transition:** Executing structured data migration strategies from legacy ERP systems to SAP S/4HANA running on high-performance database engines.
- **Invoice Automation Pipeline:** An AI/OCR-powered document processing pipeline designed for automated invoice extraction, parsing, and validation.
- **Backend Engine:** Scalable microservices built using **Python FastAPI** to orchestrate business workflows, webhooks, and system integration points.
- **Infrastructure as Code (IaC):** Automated cloud provisioning and environment lifecycle management via **Terraform** and **Azure DevOps**.
- **Enterprise Observability:** Unified monitoring, distributed tracing, and security log management using **Datadog** and **Splunk**.

2. System Architecture & Component Breakdown

2.1 Core ERP & Database Tier

- **SAP S/4HANA Platform:** Central system of record for financial, procurement, and enterprise management processes.
- **Database Management:**
 - **SAP HANA:** In-memory, column-oriented relational database powering high-speed S/4HANA transactions and real-time operational analytics.
 - **Azure Database for PostgreSQL:** Managed relational database hosting non-SAP operational data, staging schemas for migration, and backend microservice persistence layers.

2.2 Automation & AI Pipeline

- **AI/OCR Invoice Processing:**
 - Automated document ingestion from email services, storage blobs, or API endpoints.
 - High-precision Optical Character Recognition (OCR) coupled with Machine Learning models to extract fields (Vendor Details, Invoice ID, Line Items, Tax, Total Amounts).
 - Data validation engine checking extracted records against SAP master data prior to document creation.

2.3 Backend Microservices & API Gateway

- **Python & FastAPI Framework:**
 - Asynchronous processing engine for high-throughput invoice handling and integrations.
 - Auto-generated OpenAPI/Swagger documentation for REST endpoints.
- **Integrations & Webhooks:**
 - **Inbound Webhooks:** Event-driven triggers for external vendor portal actions or document state changes.
 - **Outbound Integrations:** Synchronous and asynchronous communication interfaces with SAP APIs and authentication services.

3. Cloud Infrastructure, Security & Governance

3.1 Azure Cloud Infrastructure

- Deployed inside multi-region Virtual Networks (VNETs) with segmented subnets (Public, Private, Database subnets).
- Enterprise connectivity linking PwC on-premises infrastructure with Azure using ExpressRoute / S2S VPN.

3.2 Security Governance Framework

- **Identity & Access Management (IAM):** Microsoft Entra ID (Azure AD) integration enforcing Role-Based Access Control (RBAC) and Least Privilege principles.
- **Secrets Management:** Azure Key Vault for securing DB credentials, API keys, and certificates.
- **Data Security:** Encryption at rest (AES-256) and encryption in transit (TLS 1.3) across all REST interfaces and webhooks.

4. DevOps, CI/CD & Infrastructure as Code

- **Infrastructure as Code (IaC):** Terraform scripts for repeatable provisioning across Sandbox, Staging, and Production environments.
- **CI/CD Pipelines:**

- Automated unit and integration testing for Python/FastAPI services.
- Static code checks (SonarQube) and security vulnerability scans.
- Multi-stage deployment workflows with gated approvals for production environments.

5. Monitoring & Observability

Tool	Primary Focus	Scope & Capabilities
Datadog	APM & Performance Metrics	Distributed tracing for FastAPI services, server resource usage (CPU/RAM/IOPS), and real-time alerting.
Splunk	Log Analytics & SIEM	Centralized audit logging, security event logging, SAP application log ingestion, and compliance reporting.

6. Operational Handover & Shadow Plan

The operational transition follows a 4-stage governance plan:

Phase	Timeline	Details & Key Milestones
Phase 1: Knowledge Acquisition (KA)	Days 1 – 15	15 Technical Sessions covering SAP S/4HANA, Azure, FastAPI, OCR pipeline, DevOps, and Governance.
Phase 2: Evaluation Window	Days 16 – 65	Mandatory 50-day final assessment window allocated by Manager.
Phase 3: Shadow Resourcing (SR)	Days 66 – 79	2 weeks practical shadow experience. Must resolve exactly 10 Jira tickets and handle live incidents.
Phase 4: Lead Resourcing (LR)	Day 80+	Independently leading the project forward with complete ownership.

Operational Mandates

1. **Jira Resolution Target:** Mandatory completion of **10 Jira tickets** during Phase 3 under peer review.
2. **Live Operations:** Participation in operational support calls and incident management during the Shadow phase.
3. **Full Handover:** Complete ownership transition on **Day 80**.